Name:______________________________________ Seat no.________

Linear Algebra (MTS203)

Midterm Exam, (30 Marks — 90 mins)

Class no. 11154 & 11155 (Fall 2018)

Exam Date - October 20, 2018

IBA Institute of Business Administration Karachi
Leadership and Ideas for Tomorrow

Question - 1 (10 marks)

(In your answer booklets), write T for true or F for false, as a response to the following statements (against the statement number). Reason is not required (or appreciated). You MUST NOT copy the statements to your answer booklet. In the following O represents the zero matrix, and I the identity matrix.

(a) Every homogeneous linear system is consistent.

(b) An $n \times n$ matrix A is nonsingular if and only if the reduced row echelon form of A is I.

(c) If A and B are nonsingular $n \times n$ matrices, then $A + B$ is also nonsingular and $(A+B)^{-1} = A^{-1} + B^{-1}$.

(d) If $A = A^{-1}$, then A must equal either I or $-I$.

(e) If A and B are $n \times n$ matrices, then $(A-B)^2 = A^2 - 2AB + B^2$.

(f) If $AB = AC$, and $A \neq O$, then $B = C$.

(g) If $AB = O$, then $BA = O$.

(h) If E is an elementary matrix, then E^T is also an elementary matrix.

(i) The product of two elementary matrices is an elementary matrix.

(j) If $\boldsymbol{x}$ and $\boldsymbol{y}$ are nonzero vectors in $\mathbb{R}^n$ and $A = \boldsymbol{x}\boldsymbol{y}^T$, then the row echelon form of A will have exactly one nonzero row.

Question - 2 (10 marks)

In the following O represents the zero matrix, and I the identity matrix. Moreover, matrices U and L are in block triangular forms as indicated, while D is in block diagonal form.

(a) Given:

$$A = \begin{bmatrix} I & O & O \\ O & I & O \\ O & B & I \end{bmatrix}$$

where all the sub-matrices are $n \times n$, determine the block form of A^{-1}.

(b) Let A be a nonsingular matrix that appears in block 2×2 form, such that the diagonal blocks A_{11} and A_{22} are square matrices of the same size. Let a block factoring of A be given as $A = U\ D\ L$ as follows:

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} = \begin{pmatrix} I & X \\ O & I \end{pmatrix} \begin{pmatrix} A_{11} - A_{12}A_{22}^{-1}A_{21} & O \\ O & A_{22} \end{pmatrix} \begin{pmatrix} I & O \\ Y & I \end{pmatrix}$$

Find X and Y in terms of the sub-matrix blocks of A.

Question - 3 (10 marks)

(a) Given:

$$R = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$$

show that R is nonsingular and $R^{-1} = R^T$.

(b) Let $\boldsymbol{u}$ be a unit vector in $\mathbb{R}^n$ (i.e., $\boldsymbol{u}^T\boldsymbol{u} = 1$), and let $H = 1 - 2\boldsymbol{u}\boldsymbol{u}^T$. Show that $H^2 = I$ (where I represents the identity matrix).

Question - 4 (10 marks)

(a) Show that if $\det(A) = 1$, then

$$\operatorname{adj}(\operatorname{adj} A) = A$$

(b) Suppose that Q is a matrix with the property $Q^{-1} = Q^T$. Show that:

$$q_{ij} = \frac{Q_{ij}}{\det(Q)}$$

Question - 5 (10 marks)

Define a vector space. Specify the eight (08) axioms of a vector space.